

THE EVOLUTION OF

Artificial Intelligence

1950 – Present

Milestones, Winters, and Breakthroughs

ERA 01 • 1950 – 1969

Foundations

The theoretical bedrock. Turing, Shannon, McCarthy, and Rosenblatt lay the groundwork for the entire field.

1950

Turing Test Proposed

Alan Turing publishes "Computing Machinery and Intelligence," proposing the first framework for evaluating whether a machine can exhibit intelligent behavior indistinguishable from a human.

1950

Theseus the Mouse

Claude Shannon builds an electromechanical mouse that navigates a maze and remembers its route. An early demonstration that machines can learn from experience.

1956

Dartmouth Conference

McCarthy, Minsky, Shannon, and Rochester officially coin the term "Artificial Intelligence," establishing AI as a formal academic discipline.

1957 & 1966

Perceptron & ELIZA

Rosenblatt introduces the Perceptron, the first trainable neural network. A decade later, Weizenbaum builds ELIZA, the first widely recognized chatbot, using pattern matching.

The First AI Winter

CAUSE

Hype outran capability

Early AI systems could not handle real-world complexity. Computing power and available data were far too limited to meet the ambitious promises being made by researchers. In 1973, the UK's Lighthill Report concluded that AI was failing to deliver on its stated goals.

IMPACT

Funding collapsed

Government and DARPA funding collapsed. Research programs across universities and industry labs were shut down. Public and investor confidence in AI dropped sharply, and the field entered a prolonged period of stagnation lasting nearly a decade.

Expert Systems Boom & the Second Winter

1980s • BOOM

Expert Systems Commercialized

Rule-based systems using IF-THEN logic (such as DEC's XCON) were widely adopted in medicine, finance, and manufacturing. They briefly restored commercial and academic interest in AI by delivering practical value in narrow, well-defined domains.

1987 - 1993 • SECOND WINTER

Brittle and Expensive

Expert systems proved expensive to maintain, difficult to scale, and too rigid outside of their narrow domains. The specialized LISP hardware market collapsed, funding declined again, and researchers began avoiding the AI label altogether.

ERA 04 • 1997 - 2017

Deep Learning Breakthrough

Hardware finally catches up with theory. GPUs, large datasets, and cloud infrastructure unlock modern AI.

1997

Deep Blue Beats Kasparov

IBM's Deep Blue defeats world chess champion Garry Kasparov, proving that specialized computation and search algorithms can outperform human expertise in complex strategic tasks.

2006 & 2012

Hinton & AlexNet

Geoffrey Hinton advances deep learning training methods. In 2012, AlexNet (Krizhevsky, Sutskever, Hinton) wins ImageNet using Nvidia GPUs. Widely called the "Big Bang" of modern AI.

2016

AlphaGo Defeats Lee Sedol

Google DeepMind's AlphaGo combines deep learning with reinforcement learning to master Go, a game far more complex than chess and long considered a benchmark for human intuition.

2017

The Transformer

Google researchers introduce the Transformer architecture ("Attention Is All You Need"). It replaces older sequence models with attention-based processing and becomes the foundation of every modern large language model.

ERA 05 • 2022 -
PRESENT

Generative & Agentic AI

AI enters public life. Foundation models scale, multimodality becomes standard, and autonomous agents emerge.

2022

ChatGPT Launches

OpenAI releases ChatGPT, bringing generative AI into mainstream public use. It becomes the fastest-growing consumer application in history, reaching 100 million users within two months.

2023 - 2026

Foundation Model Race

OpenAI, Google DeepMind, Anthropic, Meta, and Nvidia lead the development of increasingly capable multimodal models spanning text, image, audio, video, and code.

2024 - 2026

Agentic AI Emerges

AI systems move beyond content generation to plan, reason, use external tools, and autonomously complete multi-step workflows across research, coding, and business tasks.

Present

Still Narrow AI

Today's systems are highly capable but still task-specific. Artificial General Intelligence (AGI), a system with human-level reasoning across any domain, has not yet been achieved.

Who Built Modern AI

PIONEERING FIGURES

Alan Turing

Proposed the Turing Test, laying the philosophical groundwork for AI.

Claude Shannon

Built Theseus and helped establish the mathematical foundations of computing.

John McCarthy

Coined "Artificial Intelligence" at Dartmouth in 1956.

Frank Rosenblatt

Created the Perceptron, the first trainable neural network.

Geoffrey Hinton

The "Godfather of AI." Advanced backpropagation and deep learning.

Krizhevsky & Sutskever

Co-created AlexNet, igniting the deep learning revolution.

LEADING COMPANIES & LABS

OpenAI

Creator of ChatGPT and GPT model family. Sparked the generative AI era.

Google DeepMind

AlphaGo, Transformer architecture, and the Gemini model family.

Anthropic

Founded on AI safety research. Creator of the Claude model family.

Nvidia

Provides the GPUs powering ~92% of global AI training and inference.

Meta

Advanced open-weight models (LLaMA) driving open-source AI adoption.

IBM

Deep Blue and Watson: early demonstrations of AI mastering human tasks.

References

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3. Roser, M. (2022, December 6). The brief history of artificial intelligence: The world has changed fast – what might be next? Our World in Data. <https://ourworldindata.org/brief-history-of-ai>